

# Yash Patel

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## Profile

Results-driven Full Stack Software Engineer with expertise in modern JavaScript frameworks (Vue.js, Next.js, Angular) and robust backend architectures (Python, .NET Core, NestJS). Experienced in designing and architecting AI-powered search systems, offline-first Progressive Web Apps, and secure real-time communication systems. Adept at taking complex business requirements from concept to highly performant deployments.

## Skills

### Programming Languages

C#, JavaScript (ES6+), TypeScript, Java, Python, SQL

### Frontend

Vue.js (2/3), React.js, Angular, Remix, Next.js, HTML5/CSS3, Tailwind

### Backend

.NET Core Web API, Node.js, Express.js, FastAPI

### Databases & Cloud

SQL Server, PostgreSQL, MongoDB, Azure Functions

## Education

07/2020 – 07/2024  
Ahmedabad  
**B.E. (Computer Engineering)**  
*L.D. College of Engineering*  
CGPA - 8.70/10

## Professional Experience

01/2024 – Present  
Ahmedabad  
**Software Engineer**  
*TatvaSoft*

- **Central Employee Management System:**
  - Spearheaded the complete migration of a large-scale application from Vue 2 to Vue 3. Refactored the entire codebase to leverage the Composition API, resulting in improved maintainability and a 50% reduction in load times.
- **Farm Harvest & Pack Management (PWA):**
  - Architected an offline-first PWA using Vue 3, Pinia, and IndexedDB. Implemented an "outbox" pattern to queue requests during connectivity loss, ensuring 100% data integrity upon syncing.
  - Designed a live data synchronization module using **SignalR** to push real-time updates on harvest metrics and inventory changes to all active field devices.
  - Implemented granular **Role-Based Access Control (RBAC)** to securely manage user permissions and restrict access to sensitive operational data across different staff levels.
- **Recruitment Management System:** Delivered a full-stack solution (.NET Core, Angular) and implemented an Azure Function to automate email notifications and file parsing, reducing manual effort by ~80%.
- Integrated frontend with backend REST APIs, establishing strict data contracts to eliminate communication errors between modules.

## Projects

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### AI-Powered Personal Knowledge Hub [↗](#)

*Next.js, NestJS, Python (FastAPI), MongoDB*

- **Full-Stack Architecture:** Implemented a comprehensive knowledge management application using a Next.js frontend and a NestJS backend API, enabling users to manage rich text notes and file attachments seamlessly.
- **AI Search Engine:** Developed a dedicated Python-based search backend using FastAPI that implements a custom Retrieval-Augmented Generation (RAG) pipeline to answer natural language queries based entirely on the user's personal data.
- **Automated Data Ingestion:** Built an asynchronous pipeline to process uploaded files, parse HTML, extract text from PDFs, and ran OCR on images to automatically generate and store vector embeddings.
- **Semantic Vector Search:** Implemented cosine similarity search over stored chunk embeddings to retrieve highly relevant context, ensuring answers are grounded with exact source references while maintaining strict data privacy.

### CareConnect - Hospital Management System [↗](#)

*.NET Core MVC, SignalR, PostgreSQL, JWT*

- **Real-Time Communication Infrastructure:** Engineered a low-latency chat module using **SignalR**, supporting 1:1 and group messaging with live "typing" status and read receipts (Sent/Delivered/Read).
- **Secure Architecture:** Implemented a granular role-based access control (RBAC) system for Admins, Physicians, and Patients using **JWT authentication** to ensure strict data privacy and HIPAA-style compliance.
- **Database Optimization:** Designed a normalized **PostgreSQL** schema to manage complex medical records, utilizing indexing strategies to reduce query execution time for patient history reports.
- **Document Handling:** Built an end-to-end file management system for securely uploading, storing, and retrieving medical reports and prescriptions.